

GENERAL PURPOSE TIMERS

General Purpose timers (GP Timer) have four modes - a simple PWM mode, upcounter, down counter and up-down counter, the required mode is selected through GPTMODE bits in GPTCONTROL register.

In Counter mode, either up, down or up-down counting is done and the interrupts are raised during overflow and underflow according to the interrupt enable bits in GPTCONTROL register. The Counter reference value is given through the GPTCOUNTREF register and the present value of the counter can be obtained through GPTCOUNTER register. The input to the counter can be an internal clock or an external clock with prescaler. GPTREPEATCOUNT register hold the number of times the counter has repeated the count if it is set to continuous count in GPTCONTROL register.

GPTCAPTURE register holds the timestamp of input. The register is updated with GPTCOUNTER value when an input is received or whenever there is a change in input.

In PWM mode, according to the duty cycle and the period, PWM-waveform is generated and the interrupts are provided at compare (falling edge) and period (rising edge) according to the interrupt enable bits in GPTCONTROL register. Duty cycle of the pulse is set in GPTCOMPARE register and the period is set in GPTCOUNTREF register. The input to PWM mode is internal or external clock with prescaler. PWM mode here will not be used to produce complementary outputs hence dead-band frequency features are not included.

Internal or external clock speed can be decreased by providing appropriate value to the GPTPRESCALE bits in GPTCLKCONTROL register.

GPTCOUNTREF and GPTCOMPARE registers are double-buffered.

GPTINPUTQUAL register holds the number of input qualification cycles for the input capture.

Following are the registers used in GP Timer peripheral:

Register name	Offset address	Accessible Size	Description
GPTCONTROL	'h00	16 bits	General Purpose Timer Control Register (16 bits)(Read and Write)
GPTCLKCONTROL	'h04	32 bits	General Purpose Timer Clock Control Register (32 bits)(Read and Write)
GPTCOUNTER	'h08	32 bits	General Purpose Timer Counter Register (32 bits)(Read Only)
GPTREPEATCOUNT	'h0C	32 bits	General Purpose Timer Repeat Count Register(32 bits)(Read Only)
GPTCOMPARE	'h10	32 bits	General Purpose Timer Compare Register (32 bits)(Read and Write)
GPTCOUNTREF	'h14	32 bits	General Purpose Timer Counter Reference Register (32 bits)(Read and Write)
GPTCAPTURE	'h18	32 bits	General Purpose Timer Capture Register(32 bits)(Read Only)

GPTINPUTQUAL	'h1C	8 bits	General Purpose Timer Input Qualification Register(4 bits)(Read and Write)
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GPTCONTROL Register

15		14		13	
GPTINPUTCAPTURE		GPTCNTR UNDERFLOW		GPTCNTR OVERFLOW	
12	11	10	9	8	7
GPT PWM RISE	GPT PWM FALL	GPTCNTR UNDERFLOW INTREN	GPTCNTR OVERFLOW INTREN	GPT PWMRISE INTREN	GPT PWMFALL INTREN
6	5	4	3	2	1
GPTCONT COUNT	GPTCOUNT RESET	GPTOUTEN	GPTMODE	GPTRESET	GPTENABLE

Bits 15 - GPTINPUTCAPTURE : Polarity of the input to be captured. When set to 1, the capture input when gone high will be captures else input when gone low will be captured

Bit 14 - GPTCNTR UNDERFLOW : read-only bit. This is set when there is an underflow from down-counter and up-down counter.

Bit 13 - GPTCNTR OVERFLOW : read-only bit. This is set when there is an overflow from up-counter and up-down counter

Bit 12 - GPTPWM RISE : read-only bit. This is set when the PWM period completes - Rising edge of the PWM mode

Bit 11 - GPTPWM FALL : read-only bit. This is set when the count value is greater than compare in PWM mode. This is falling edge of PWM signal

Bit 10 - GPTCNTR UNDERFLOW INTREN : This bit should be set if interrupt is needed during counter underflow

Bit 9 - GPTCNTR OVERFLOW INTREN : This bits should be set if interrupt is needed during counter overflow

Bit 8 - GPT PWMRISE INTREN : This bit should be set if interrupt is needed at PWM Rising edge

Bit 7 - GPT PWMFALL INTREN : This bit should be set if interrupt is needed at PWM Falling edge

Bit 6 - GPTCONT COUNT : When this bit is set, the counter will be in continuous count mode.

Bit 5 - GPTCOUNT RESET : This bit can be set to reset the counter register in the GPTimer

Bit 4 - GPTOUTEN : When this bit is set, PWM output will be made available at the GPTimer output

Bit 3:2 - GPTMODE : These bits define the mode of GPTimer. 00 - PWM; 01 - Up counter; 10 - Down counter; 11 - Up-Down counter

Bit 1 - GPTRESET : This bit is to reset the GP Timer. This when set(1), resets the clocks and registers which are used for counting or PWM operation
 Bit 0 - GPTENABLE : This bit is set(1) to enable GP Timer and reset(0) to disable

GPTCLKCONTROL Register

31	17 16	1	0
Reserved		GPTPRESCALE	GPTCLKSRC

Bits 31:17 - Reserved
 Bits 16:1 - GPTPRESCALE : Prescale value to get required clock speed.
 Required clock frequency = Internal clock frequency / (Prescale + 1).
 Bit 0 - GPTCLKSRC : GP Timer Clock source. This bit when set(1), external clock is used and when reset(0), internal clock is used.

Sequence of execution:

1. Write into compare, counterref, input qualification register and clock control register
2. Write into the control register to start the counter operation
3. Read corresponding registers such as counter register for up/down count values, capture register to get the timestamp of the input and repeat count register to get the number of times the counter has repeated the counting